

EDUGENIE

Phase 1 – Brainstorming & Ideation

AI-Powered Learning Assistant

1. Introduction

EduGenie is a Generative AI-based learning assistant designed to support students with common academic learning activities. The brainstorming phase focuses on identifying student learning challenges, possible AI-based solutions, useful features, and a practical project scope for an educational application.

2. Problem Identification

Students may need help with understanding academic concepts, obtaining concise answers, summarizing study material, preparing practice questions, and deciding what to learn next. The project idea is to combine several of these study-support activities into one simple AI-powered application.

3. Brainstormed Problems

- Difficulty getting quick and understandable answers to academic questions.
- Complex topics may require simpler, step-by-step explanations.
- Long study material can be time-consuming to review.
- Preparing practice quizzes manually can take additional time.
- Students may need guidance on what topics to study next.
- Using separate tools for different study activities can be inconvenient.

4. Brainstormed Solution Ideas

Idea / Feature	Purpose
AI Question & Answer	Allow students to ask academic questions and receive AI-generated answers.
Topic Explanation	Provide simplified explanations of difficult concepts.
Text Summarization	Convert longer learning content into concise summaries.
Quiz Generation	Generate practice multiple-choice questions from a topic or learning content.
Learning Recommendations	Suggest relevant learning activities or next topics based on student input.

5. Selected Project Idea

The selected idea is EduGenie – an AI-powered learning assistant that combines multiple study-support functions in one application.

- Question and Answer
- Concept Explanation
- Text Summarization
- Quiz Generation
- Learning Recommendations

6. Purpose of the Project

The purpose of EduGenie is to provide students with an accessible AI-based study assistant that supports understanding, revision, practice, and learning guidance through a single application.

7. Target Users

- College and university students.
- Learners who need quick explanations of academic concepts.
- Students preparing for revision and self-practice.
- Learners who want AI-assisted study guidance.

8. Key Benefits Identified During Brainstorming

- Provides multiple learning-support functions in one application.
- Supports interactive question-and-answer based learning.
- Can simplify difficult concepts into understandable explanations.
- Helps students create concise study summaries.
- Provides automatically generated practice quizzes.
- Can provide learning recommendations to support continued study.

9. Initial Technology Direction

The brainstorming phase identified a web-based application using Python and Generative AI as a practical implementation direction.

- Python
- FastAPI
- Google Gemini Generative AI
- HTML, CSS and JavaScript
- Pydantic
- Pytest for testing

10. Initial Project Scope

The initial scope is to develop a functional prototype containing the five selected learning features. The application should accept student input, process the request using the AI service, and present the generated result through a simple web interface.

11. Brainstorming Outcome

The brainstorming and ideation phase resulted in a clear project concept: EduGenie will serve as a Generative AI-powered learning assistant with features for Q&A, explanations, summarization, quiz generation, and learning recommendations. This provides the foundation for the Requirement Analysis phase.

12. Conclusion

The brainstorming phase helped identify major learning challenges and convert them into a practical software idea. EduGenie was selected as the project concept because its proposed features combine several common study-support activities into one AI-assisted learning application.